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Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?



The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does "qualifying text" mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc.

Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

Comparative Fake News Detection Using Machine Learning and Deep Learning Models

Student Name:

Student Id:

1.Project Overview

At present, due to easy access to digital platforms, misinformation is being spread rapidly. Never before has the average user been exposed to such false, incorrect information. This is the place where fake news detection is essential. Stakeholders are influenced in the wrong manner, as are the general public due to misinformation that is being circulated. An automated system capable of quickly addressing the issue of misinformation would be very valuable. Therefore, it becomes very essential to do a comparative study of the machines learning method and deep learning approach.

This project studies using fake news detection with a traditional, neural network and transformer-based model. A Logistic Regression model is the traditional model. The model a Bidirectional Long Short-Term Memory Model is a neural network model.

The aim of this project is to assess and contrast how the various models perform on a fake news dataset with classifying the news as real or fake.

2. Research Question

How do traditional machine learning models compare with deep learning and transformer-based models in detecting fake news, in terms of accuracy, efficiency, and practical usability?

3. Objectives

- To develop a fake news classification system using multiple approaches.
- To implement and compare Logistic Regression, BiLSTM, and DistilBERT models.
- To evaluate model performance using accuracy, precision, recall, and F1-score.
- To study how predictive power changes with varying algorithm complexity and size.
- To deploy the models through a REST API and web-based dashboard for real-time interaction.

4. Methodology

In this project, models in the machine learning pipeline database will have a systematic development and evaluation process.

4.1 Data Collection

We will use a popular dataset which contains articles of fake news and real news. The labeled data contains a large number of examples to be sufficient.

4.2 Data Preprocessing

The text data will be cleansed and standardised through tokenisation, lowercasing, stop words removal, removal of additional symbols, etc. Via TF-IDF vectorisation, we will feed traditional models. We will use tokenized sequences and embeddings on the deep learning models.

4.3 Model Development

We are going to implement three models. To accomplish this task we used three methods namely logistic regression, which is our baseline model, BiLSTMs to learn a sequential pattern from a sentence and the DistilBERT to take advantage of contextual understanding.

4.4 Training and Evaluation

By using sentiment analysis, machine learning algorithms will be deployed. The analysis will identify feelings of joy, sad, annoyed and other emotions. Four models with their respective sub-models will be evaluated.

4.5 Deployment

Flask will be used to create the REST API. Even though we don’t need a highly sophisticated web-interface, we will build a very rudimentary interface where a user can input the news text and see the prediction of all the models for.

5. Expected Deliverables

I created machine learning and deep learning models for fake news classification.

- A comparison of model performance using evaluation metrics.
- A REST API for real-time fake news detection.
- A dashboard on the web for interacting with the model and making comparisons
- Graphical representation such as confusion matrix and performance comparison plot.

6. Resources Required

The project will require the following resources:

- Programming Language: Python
- Libraries: Scikit-learn, TensorFlow or PyTorch, HuggingFace Transformers, NLTK
- Tools: Flask for API development, Jupyter Notebook for experimentation
- Hardware: A standard personal computer, with optional GPU support for training deep learning models

7. Ethical Considerations

The data used for this project will be in the public domain and does not consist of any personal or sensitive data. However, biases in datasets that may affect the fairness and accuracy of a model raise ethical concerns. When implemented, a wrong classification of news articles may have real-world consequences. As a result, the project will understand the limitations of the dataset and use the system responsibly. Users will receive counseling about avoiding misleading predictions generated.

8. Critical Evaluation and Limitations

Even though it is anticipated that the models that are implemented are going to delivery a high performance, it is critical to analyse the weaknesses. The dataset used in this story may not adequately reflect the complex world of news, leading to overly optimistic results. Overfitting is possible, particularly in deep learning and models based on transformers. Moreover, higher-level models like DistilBERT have almost no extra predictive power but use a lot more computation. Through this project, we will look at these tradeoffs and how performance may differ from the model and ways to mitigate tradeoffs.

9. Conclusion

The purpose of this project is to do a comparative analysis of machine learning and deep learning techniques for fake news detection. The models will be implemented and evaluated in order to showcase their strengths and weaknesses, in terms of accuracy, efficiency and usability. Putting a system to deploy shows a really practical utility. Further, it is through the critical assessment that application finding becomes real-world-based. The project's overall goal is to increase the reliability and effectiveness of automated tools that detect misinformation.